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Type 2 diabetes mellitus patients manifest characteristic spatial EMG potential distribution pattern during sustained isometric contraction

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ABSTRACT

Aim: The purpose of the present study is to investigate spatial surface electromyography (SEMG) potential distribution pattern in type 2 diabetes mellitus (T2DM) patients.
Methods: Nine T2DM patients and nine age-matched healthy men (CON) performed a sustained isometric knee extension at 10% of maximal voluntary contraction for 120 s. Multi-channel SEMG was recorded from the vastus lateralis muscle by means of 64 electrodes. To characterize spatial SEMG potential distribution pattern, modified entropy and correlation coefficients between same electrode locations were calculated at 15, 60 and 120 s for the root mean square values.
Results: At 60 and 120 s, modified entropy in T2DM was significantly lower than those in CON ($p < 0.05$). Correlation coefficients for T2DM were significantly higher than those for CON at 60 and 120 s ($p < 0.05$).
Conclusion: From these results, we suggested that T2DM patients continue to recruit limited and same motor units during the sustained contraction at low force level.
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1. Introduction

Recent estimates indicate there were 171 million people in the world with diabetes in the year 2000 and this is projected to increase to 366 million by 2030 [1]. In diabetes patients, 90–95% of cases were categorized as type 2 diabetes mellitus (T2DM), previously referred as non-insulin dependent diabetes. This type of diabetes encompasses individuals who have insulin resistance and usually have relative insulin deficiency. More importantly, T2DM is a significant cause of premature mortality and morbidity related to cardiovascular disease, blindness, kidney and nerve disease, and amputation [2].

For the prevention and management of T2DM, exercise has been strongly recommended along with diet and medication [3]. The previous studies have collected physiological responses to exercise in T2DM patients in order to understand physiological characteristics during exercise and design effective exercise program for this type of patients [3]. Since

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Abbreviations: CON, control; MF, median frequency; MU, motor unit; MVC, maximal voluntary contractions; RMS, root mean square; SEMG, surface electromyography; T2DM, type 2 diabetes mellitus; VL, vastus lateralis.
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T2DM is caused by dysfunction in metabolic system, most of the previous studies had focused on metabolic and cardiovascular responses to exercise in T2DM patients [4–8]. These studies also demonstrated the premature muscular fatigue and the reduced exercise tolerance in T2DM patient during treadmill exercise, cycling, and planter flexion exercise [4–7].

On the other hand, functional impairments in neuromuscular system had also been observed. Hyperglycemia in T2DM patients induces oxidative stress in diabetic neurons and results in diabetic neuropathy, i.e. diabetic distal symmetrical sensorimotor polyneuropathy and diabetic automatic neuropathy [9–11]. Effect of diabetic neuropathy on motor control during quiet standing and gait had been reported [12–15]. Central nervous system is regulated by afferent input from the receptors connected with peripheral nervous in even simple muscle contraction at a joint as well as in other complex human movement [16]. We, thus, suspect that some degree of neuromuscular impairments might be present in those patients with T2DM during sustained muscle contraction in which both neural and metabolic adjustments would be required. In addition to metabolic dysfunction, neuromuscular system also may be one of the factors of the premature muscular fatigue and the reduced exercise tolerance in T2DM patient [4–6].

Recently, neuromuscular functions such as motor unit (MU) recruitment strategy or functional compartmentalization within a muscle have been assessed from spatial distribution pattern of muscle activation by using multichannel surface electromyography (SEMG) technique [17–22]. The previous studies demonstrated that spatial EMG potential distribution pattern within a muscle is altered by contraction levels or fatigue [17,21–24]. This phenomenon has been explained by a spatial inhomogeneity in the location of different types of muscle fibers [25] and a clustering of muscle fiber innervated by one MU in limited territory [26]. Recruitment and rate coding for these different types of muscle fibers for increasing torque or fatigue would induce changes in spatial distribution of SEMG potential. For sustained contraction, nociceptive afferent input from contracted muscle to central nervous system was suggested as one of the major mechanisms of alteration in spatial distribution (redistribution) of muscle activation [18,19,27]. Assessment of spatial distribution pattern of muscle activation would be an efficient tool to investigate neuromuscular function related with the responses in peripheral nervous system.

The purpose of the present study is to investigate spatial distribution pattern of muscle activation during sustained contraction in T2DM patients. We hypothesized that redistribution of spatial EMG potential distribution pattern is attenuated in T2DM patients as some degree of neuromuscular impairments might be present in those patients during sustained muscle contraction in which both neural and metabolic adjustments would be required.

2. Materials and methods

2.1. Subjects

Nine elderly men with type 2 diabetes mellitus (T2DM) and nine age-matched healthy men (CON) participated in this study. All subjects in T2DM group have been diagnosed as T2DM and treated in the hospital for 7–38 years (Table 1). The subjects of both groups gave written informed consent for the study after receiving a detailed explanation of the purposes, potential benefits, and risks associated with participation in the study. Age, body mass, BMI, maximal voluntary contraction (MVC) torque during isometric knee extension and MVC torque relative to body mass were matched between the groups (Table 1). All subjects in both groups had no history of any locomotor disorders. Blood sample was collected to determine the concentration of glycosylated hemoglobin (HbA1C) levels, which is used as an index of average blood glucose levels over the preceding 2–3 months and as a diagnostic criteria for diabetes mellitus [28]. All procedures used in this study were in accordance with the Declaration of Helsinki and were approved by the Committee for Human Experimentation at the Graduate School of Human and Environmental Studies, Kyoto University and for Kyoto Teishin Hospital.

2.2. Experimental design

The subjects were tested for maximal voluntary contractions (MVC) during isometric knee extension according to our previous procedures [21,22]. After sufficient rest period, each subject performed a sustained contraction at 10% of MVC for 120 s during isometric knee extension. During sustained contraction, multi-channel SEMG was recorded from the vastus lateralis (VL) muscle.

Isometric knee extensions were performed on a custom dynamometer mounting a force transducer (LU-100KSE; Kyowa Electronic Instruments, Tokyo, Japan). During contraction, both hip and knee joint angles were flexed at 90° (180° is fully extended), respectively. The MVC involved a gradual increase in knee extension force exerted by the knee extensor muscles from baseline to maximum in 2–3 s and then sustained at maximum for 2 s. The timing of the task was based on a verbal count given at a 1-s interval, with vigorous encouragement from the investigators when the force began to plateau. The subjects performed at least two MVC trials with ≥2 min rest between trials. The highest MVC force was used to calculate the MVC torque and target torque for sustained

Table 1 – Characteristics of the subjects for type 2 diabetes mellitus patients (T2DM) and age-matched healthy control (CON).

	T2DM	CON
Age (year)	70.6 ± 6.7	72.6 ± 3.8
Height (cm)	166.0 ± 7.0	163.8 ± 3.2
Body mass (kg)	61.9 ± 7.3	62.9 ± 3.6
BMI	22.5 ± 2.3	23.4 ± 1.5
MVC (N m)	116.7 ± 19.8	124.0 ± 29.6
MVC/body mass (N m/kg)	1.9 ± 0.3	2.0 ± 0.5
HbA1C (%)	7.9 ± 0.9	5.3 ± 1.7*
Duration of T2DM (year)	18.9 ± 11.9	
Subcutaneous tissue thickness (cm)	0.36 ± 0.18	0.40 ± 0.07
Muscle thickness (cm)	2.03 ± 0.31	2.21 ± 0.33

Data are mean ± SD.
* $p < 0.05$ vs. T2DM.

contraction. Knee extension torque was calculated as the product of the knee extension force and length between the estimated knee joint center and the distal portion of the shank linked to force transducer. After MVC, the sustained contraction at 10% of MVC force was performed for 120 s. The produced and target torques were shown to the subjects on a personal computer monitor. Subjects practiced MVC and sustained contraction ≥10 min before test session.

2.3. EMG recording

Multi-channel SEMG signals were detected from the VL muscle with a semi-disposable adhesive grid of 64 electrodes (ELSCH064R3S, OT Bioelectronica, Torino, Italy) using the same procedure as that used in our previous study [21,22]. This muscle is one of the knee extensor muscles which play important roles during human movements and most previous studies have focused on this muscle to investigate disease related changes in metabolism or histochemistry of a skeletal muscle [7,8,29]. Thus, we selected the VL muscle to detect SEMG in the present study. The grid is made of 13 rows and 5 columns of electrodes (1 mm diameter, 8 mm inter-electrode distance in both directions) with one missing electrode at the upper left corner. Prior to attaching the electrode grid, the skin was cleaned with alcohol. Conductive gels were inserted into the cavities of the grid electrode to assure proper electrode skin contact. The center of electrode grid was placed at midpoint of the line between the head of great trochanter and inferior lateral edge of patella. The rows of electrodes were placed along the longitudinal axis of VL muscle such as the line between the head of great trochanter and inferior lateral edge of patella. The position of missing electrode was located at proximal side of longitudinal axis of VL muscle. The grid electrode was connected to the amplifier through 4 connectors which were fixed at the subject skin by elastic tape. A reference electrode was placed at the iliac crest. At the center of electrode location, longitudinal ultrasonographic image (SSD-900, ALOKA, Tokyo, Japan) was taken to determine the thickness of the subcutaneous tissue and VL muscle.

Monopolar SEMG signals were amplified by a factor of 1000, sampled at 2048 Hz and converted to digital form by a 12-bit analog-to-digital converter (EMG-USB, OT Bioelectronica, Torino, Italy) with the signal of force transducer. Recorded monopolar SEMG signals were off-line band-pass filtered (10–500 Hz) and transferred into analysis software (MATLAB R2009b, MathWorks GK, Tokyo, Japan). Fifty-nine bipolar SEMG signals along the rows were made from 64 electrodes. To calculate root mean square (RMS) and median frequency (MF), SEMG signals were sampled over 1 s from 1 s before the given time to the given time at 15 s, 60 s, and 120 s. From 59 RMS and MF values normalized by the value at 15 s, mean normalized RMS and MF values were calculated at 60 s and 120 s.

Modified entropy was calculated for 59 absolute RMS values (in space) at each time as done by Farina in a previous work [17]. Decrease in modified entropy means that increase of heterogeneity in spatial EMG potential distribution within an electrode grid.

In each time point, 59 absolute RMS values were categorized into three activation level by the percentage of peak RMS value at each time, i.e. low (0–33% of peak RMS), middle (33–66% of peak RMS), and high (66–100% of peak RMS) activation. Number of channel was counted in individual activation levels.

To characterize changes in spatial SEMG potential distribution with time course, correlation coefficients were calculated from the 59 pairs of absolute RMS values (RMS map) at same locations at between 15 s and other two sampled time.

2.4. Statistics

All data are provided as mean and SD. Before the analysis, the normal distribution of the data was confirmed using Shapiro–Wilk test. The parametric analysis was used for normally distributed data and the non-parametric analysis was used for non-normally distributed data. Age, body mass, BMI, MVC torque, MVC torque relative to body mass, HbA1C, and VL muscle thickness were compared between groups using *t*-test. Thickness of subcutaneous tissue was compared between groups using Mann–Whitney *U*-test. Friedman test and Mann–Whitney *U*-test were performed for mean normalized RMS and MF values, number of channel with three activation levels, modified entropy, correlation coefficient of RMS map, and performed force to investigate changes with time course and to compare between the groups at the given times, respectively. The level of statistical significance was set at $p < 0.05$. Statistical analyses were performed using SPSS software (version 15.0; SPSS, Tokyo, Japan).

3. Results

There were no significant differences between the groups in anthropometric parameters, MVC torque, MVC torque relative to body mass, and thickness of subcutaneous tissue and VL muscle ($p > 0.05$) (Table 1). A significant difference between the groups was observed in HbA1C as expected ($p < 0.05$) (Table 1).

For mean normalized RMS and MF values, there were no significant changes with time course in both groups and no significant difference between the groups ($p > 0.05$). There were also no significant differences in the performed forces at the given times ($p > 0.05$), indicating that the targeted torque was well controlled by the subjects of both groups.

Fig. 1 illustrated representative multi-channel SEMG amplitude shown as color map at the given times for T2DM and CON groups. In T2DM, large areas with low RMS value were demonstrated at all times. Changes in spatial SEMG potential distribution with time course were seen in both groups. However, changes in spatial SEMG potential distribution in CON group were greater than that in T2DM group in these representative data.

No significant change with time was found in modified entropy for both groups ($p > 0.05$) (Fig. 3). Modified entropy in T2DM group was significantly lower at 60 and 120 s than those in CON group ($p < 0.05$) (Fig. 2), indicating that heterogeneity in spatial EMG potential distribution was greater in T2DM group.

There were no significant changes with time in number of channel with all activation level for both groups ($p > 0.05$).

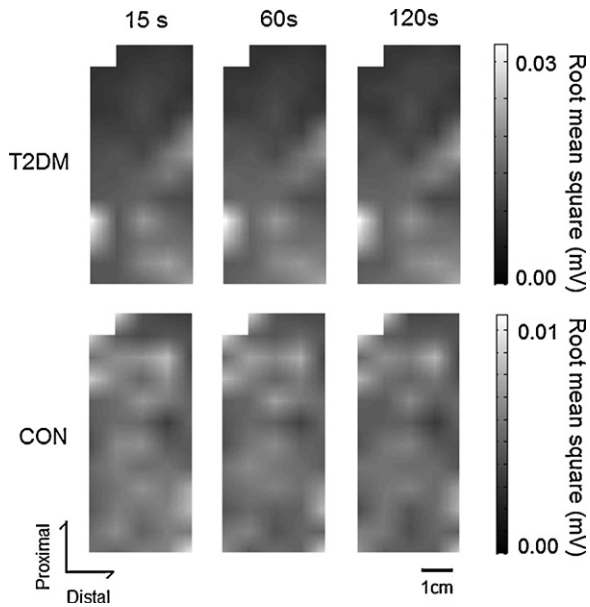

Fig. 1 – Representative root mean square value for all channels shown as color map at selected times during sustained contraction for the subjects from type 2 diabetic mellitus patient (T2DM) group and age-matched healthy control group (CON).

Thus, numbers of channel with three activation levels were shown only at 120 s in Fig. 3. Number of channel with low activation was significantly greater in T2DM group than CON group ($p < 0.05$). In T2DM group, numbers of channel with low and middle activation were significantly greater than high activation level ($p < 0.05$). On the other hand, in CON group, numbers of channel with middle activation level were significantly greater than low and high activation level ($p < 0.05$).

Significant changes were found in correlation coefficients of RMS map with time course in both groups ($p < 0.05$). At 60 s and 120 s, correlation coefficients of RMS map in T2DM were significantly higher than those in CON ($p < 0.05$) (Fig. 4). This means that the time course change in spatial SEMG potential distribution was smaller in T2DM group.

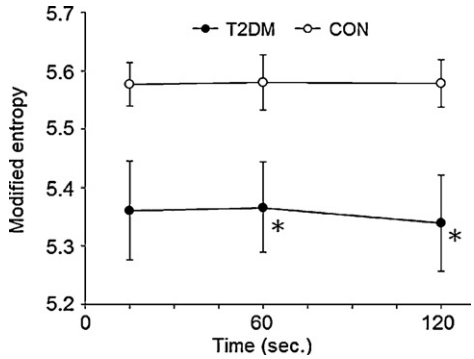

Fig. 2 – Mean (±SE) modified entropy during sustained contraction. T2DM, type 2 diabetes mellitus patients group; CON, age-matched healthy control group. *$p < 0.05$ vs. CON group.

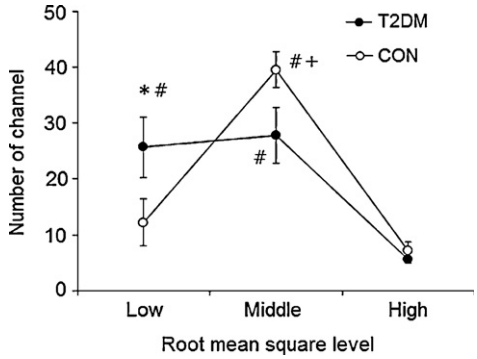

Fig. 3 – Mean (±SE) number of channel with three different root mean square levels during sustained contraction at 120 s. T2DM, type 2 diabetes mellitus patients group; CON, age-matched healthy control group. *$p < 0.05$ vs. CON group. #$p < 0.05$ vs. high level. +$p < 0.05$ vs. low level.

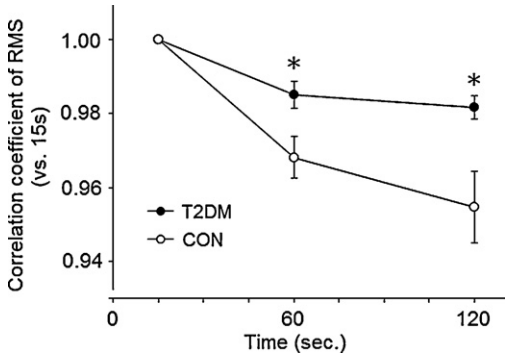

Fig. 4 – Mean (±SE) correlation coefficient values in root mean square map between at 15 s and 120 s during sustained contraction. T2DM, type 2 diabetes mellitus patients group; CON, age-matched healthy control group. *$p < 0.05$ vs. CON group.

4. Discussion

In the present study, mean normalized RMS and MF values did not change with time for both groups. During relative low-level sustained contraction (<10% of MVC), blood flow through the muscle is sufficient to prevent fatigue [30]. Thus, fatigue-induced progressive MU recruitment, increased firing frequency and decrement in conduction velocity of action potential, which are causes of increase in RMS and decrease in MF during sustained contraction [31–33], could not be occurred in both groups. Also, there were no significant differences in these global SEMG variables between the groups at the given times, although the previous reports demonstrated premature muscular fatigue in T2DM patient [4–6]. These findings would indicate that the given task in the present study does not induce muscular fatigue even in T2DM patients. On the other hand, the fatigability during isometric contraction depends on absolute force [34]. In the present study, absolute target force during sustained contraction was matched between the groups owing to matched absolute MVC torque. We thus

assumed that the burden for working muscle was controlled between the groups.

Heterogeneity in spatial EMG potential distribution within an electrode grid was greater in T2DM group in the present study (Fig. 2). This would be due to greater number of electrode with low RMS values in T2DM group as compared with CON group (Fig. 3). From these findings, it was suggested that limited area was activated within a muscle during a sustained contraction in T2DM patients. Heterogeneity in spatial EMG potential distribution can be explained by spatial inhomogeneity in the location of different types of muscle fibers [25] and a clustering of muscle fiber innervated by one MU in limited territory [26]. We thus supposed that in T2DM patients limited MUs were recruited during sustained contraction at low force level. Since the previous studies demonstrated that a reduction in slow oxidative muscle fibers or lower percentage of type 1 muscle fiber in VL muscle of T2DM patients [7,8,29], muscle fibers or MUs contributing to low level contraction may be smaller in T2DM patients. Moreover, denervation of muscle fibers and/or increase of intramuscular fat tissue caused by diabetic amyotrophy have been demonstrated in diabetes mellitus patients including T2DM patients [11]. These morphological changes may also induce heterogeneity spatial EMG potential distribution within a muscle in T2DM group.

While RMS and MF values calculated from all electrode pairs were unchanged with time, spatial distribution pattern of SEMG changed with time for both groups. Also, change in spatial distribution pattern of SEMG was smaller in T2DM group in the present study (Fig. 4). Under the assumption that the observed changes in spatial EMG potential distribution pattern might reflect recruitment of heterogeneously located MUs with limited territory within a muscle [25,26], our data suggest that limited number of the same MUs might have been activated continuously during sustained contraction in T2DM patients. Since chemical responses such as blood lactate concentration arise even in low-level sustained contraction (10% of MVC) [30], chemical stimuli may be one of the causes of change in spatial distribution pattern. It is well known that central nervous system is regulated by afferent input from the receptors within a muscle during contraction [16]. Madeleine et al. [18] and Falla et al. [27] demonstrated that nociceptive afferent input elicited by experimental muscle pain changes spatial distribution pattern of SEMG during sustained contraction in the upper trapezius muscle. These results indicate nociceptive afferent input contributes to recruitment and/or derecruitment of MU during a sustained contraction. Diabetic peripheral neuropathy is one of the severe complications in T2DM patients [9–11]. In particular, dysfunction in small diameter nerves, i.e. pain, thermal perception, and pressure, named as diabetic distal symmetrical sensorimotor polyneuropathy, is early and often occurs in T2DM patients [9–11]. We thus infer that diabetic peripheral neuropathy decreases afferent input to central nervous system during muscle contraction. Due to this reduction in afferent input, recruitment and/or derecruitment could be not progressed during a sustained contraction and thereby attenuates change in spatial distribution of muscle activation in T2DM patients. However, degree of diabetic peripheral neuropathy for T2DM patients was not assessed in the present study. More detailed work is necessary to investigate the relationship between spatial EMG potential distribution pattern and diabetic peripheral neuropathy.

Farina et al. [17] showed that changes in spatial distribution of SEMG potential correlate with exhaustion time during low level isometric contraction for the upper trapezius muscle. This suggests that redistribution of muscle activation plays a key role of prolonging muscular fatigue during sustained contraction [17]. The premature muscular fatigue and the reduced exercise tolerance in T2DM patient are well known [4–6] and it has been recognized that dysfunction in metabolic and cardiovascular systems are main causes of that [4–8]. From the result of the present study, it was assumed that in addition to dysfunction in metabolic and cardiovascular systems specific activation pattern in neuromuscular system could also contribute to premature muscular fatigue and the reduced exercise tolerance in T2DM patients.

In conclusion, we compared spatial distribution pattern of muscle activation during sustained contraction between T2DM patients and age-matched healthy men using multi-channel SEMG for the knee extensor muscle. Limited area was activated within a muscle and the attenuation of redistribution in spatial EMG potential pattern was seen in T2DM patients. From these results, we suggested that T2DM patients might activate limited numbers of the same MUs continuously during the sustained contraction at low force level.

Conflict of interest

The authors declare that they have no conflict of interest.

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